



Open-CD: A Comprehensive Toolbox for Change Detection

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Abstract

We present Open-CD, a change detection toolbox that contains a rich set of change detection methods as well as related components and modules. The toolbox started from a series of open source general vision task tools, including OpenMMLab Toolkits, PyTorch Image Models (Timm), *etc.* It gradually evolves into a unified platform that covers many popular change detection methods and contemporary modules. It not only includes training and inference codes, but also provides some useful scripts for data analysis. We believe this toolbox is by far the most comprehensive change detection toolbox.

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In this report, we introduce the features, supported methods and applications of Open-CD. In addition, we also conduct a benchmarking study on different methods and components. We wish that the toolbox and benchmark could serve the growing research community by providing a flexible toolkit to re-implement existing methods and develop their own new change detectors. Code and models are available at <https://github.com/likyoo/open-cd>.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

open source; change detection; semantic change detection

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1 Introduction

Change detection is a fundamental remote sensing image interpretation task. The pipeline of change detection frameworks is usually similar to image segmentation tasks. But in particular, there is a pair of images as input, and the task focuses on detecting pixel-level differences in the bi-temporal images. Therefore, change detection is more complicated than single-temporal segmentation. And like other high-level vision tasks, different implementation settings can lead to very different results. Towards the goal of providing a high quality codebase and unified benchmark, we build Open-CD, an open-source change detection toolkit.

Major features of Open-CD are: (1) **Reliable dependencies.** Open-CD is built on the OpenMMLab Toolkits, especially MMCV, MMEEngine, MMEPretrain, MMSEgmentation and MMDetection, and can arbitrarily call their components, including pipeline, model, module, loss, data augmentation, *etc.* in the config file. (2) **Modular design.** We decompose the change detection framework into different components and users can easily construct a customized change detection method by combining different modules. (3) **Support of multiple methods out of box.** Open-CD supports typical and popular change detection methods, see Section 2 for the full list. (4) **State of the art.** In most cases, the Open-CD implementation performs better than the official one. It also helps some users win change detection challenges. (5) **High efficiency.** Depending on efficient MMEEngine and MMCV, the training speed of Open-CD is faster than or comparable to other codebases.

In this report, we introduce the composition and workflow of the Open-CD toolbox, and conduct a benchmarking study on the typical or state-of-the-art change detection algorithms, including F_1^c score, parameter, inference speed, *etc.*

2 Supported methods

2.1 Existing Change Detection Methods

Open-CD contains high-quality implementations of popular change detection methods. A summary of supported frameworks and features compared with other codebases is provided in Table 1. In addition, we invite some authors to introduce their algorithms and participate in the construction of the Open-CD toolkit, as described at https://github.com/likyoo/open-cd/tree/main/projects/open-cd_technical_report.

2.2 Customized Change Detection Models

There is no doubt that the construction of change detection models benefits from other fundamental tasks in general vision, especially classification and segmentation. The former provides powerful feature extractors (backbones) and the latter provides extensive change mask generation schemes. Therefore, a core aspect of Open-CD's design is to allow users to arbitrarily call components from other vision models and to make it easier for change detection to benefit from advanced general modules. This is an important reason why we build Open-CD on top of OpenMMLab and Timm toolkits.

3 Architecture

3.1 Overall

As shown in Figure 1, Open-CD consists of five parts.

Config Files. In Open-CD, the model, optimizer, dataset, data pipeline, *etc.* are specified in config files. In general, modifying the config file is sufficient unless internal model redesign is required.

Model Zoo. Besides change detection models listed in Table 1, Open-CD supports almost all models in OpenMMLab toolkits, which can be called directly in the config file by simply installing them as dependencies. In addition, there are thousands of pre-trained weights available.

Internal Modules. Following the design concept of the OpenMMLab toolkits, the training process of Open-CD is conducted in the MMEEngine Runner. We will describe it in Section 3.2.

Data Files. Open-CD supports the current mainstream 9 BCD dataset and 3 SCD datasets. Since we define two extensible base classes, `_BaseCDDataset` and `BaseSCDDataset`, it is easy for users to integrate custom datasets into Open-CD.

Tools. Open-CD also contains several functional scripts *e.g.* training tools, inference tools, data analysis tools, result analysis tools, *etc.*

3.2 Training Pipeline

Open-CD follows the unified interface of abstract components defined in MMEEngine, consistent with other OpenMMLab toolkits. The training begins with the Dataloader loading data from the dataset and feeding it to the model, which is wrapped for distributed training, *etc.* A Scheduler adjusts optimizer parameters (*e.g.*, learning rate), while the Optimizer updates the model weights. The optimizer wrapper also supports advanced features like gradient accumulation, clipping, and mixed precision training. During evaluation, the data and model outputs are passed to the Evaluator, which computes and returns the relevant metrics.

To enhance flexibility, hook points are embedded throughout the pipeline, allowing users to observe or modify the training process via the Runner. Throughout the entire process, the Logging component continuously transmits information to the Visualizer.

4 Benchmarks

4.1 Experimental Setting

Dataset and Implementation details. We use LEVIR-CD as the main benchmark due to its high quality and wide adoption, training on the Train split and evaluating on the Test split. By default, images are randomly cropped to 256×256 , and models are trained and evaluated on a single RTX 3090 GPU with a batch size of 8 for 40k iterations. Data augmentation includes RandomRotate, RandomFlip, and PhotoMetricDistortion.

Evaluation metrics. We adopt F_1^c , IoU^c , $Precision^c$ and $Recall^c$ as evaluation metrics. Note that all metrics are based on the category “change” to avoid the class imbalance problem.

4.2 Results

We compare the official implementations of some methods with those in Open-CD, as shown in Table 2. Most implementations in Open-CD gain some degree of performance improvement compared to the original implementations. Especially for BIT, ChangeStar

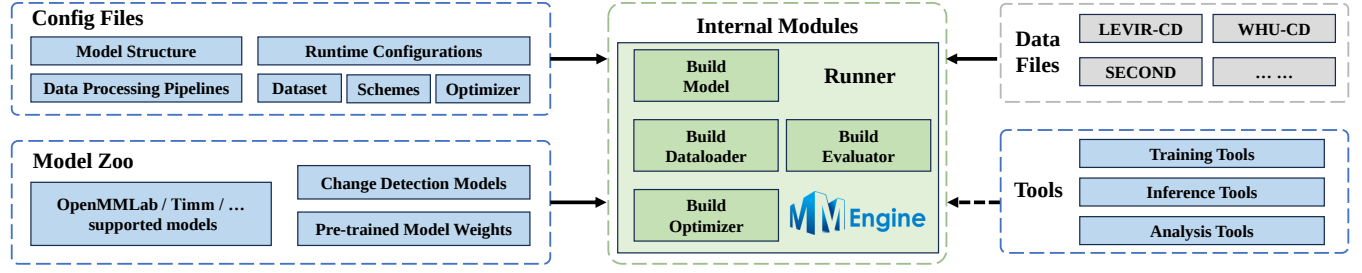


Figure 1: The overall architecture of Open-CD.

Table 1: Comparisons of different change detection codebases. “✓” means officially supported and blank means not supported.

	Open-CD PyTorch	change_detection.pytorch ¹ PyTorch	ChangeDetectionRepository ² PyTorch	CDLab ³ 飞书	ChangeDetectionToolbox ⁴ MATLAB
Tasks:					
Binary	✓	✓	✓	✓	✓
Semantic	✓				
Traditional Methods:			✓		✓
Customized Models:					
Classification Nets	✓	✓			
Segmentation Nets	✓	✓			
Specific Models:					
FC-EF [6]	✓			✓	
FC-Siam-Diff [6]	✓			✓	
FC-Siam-Conc [6]	✓			✓	
STANet [3]	✓	✓		✓	
DSIFN [20]	✓			✓	
L-UNet [14]				✓	
SNUNet [8]	✓		✓	✓	
BIT [2]	✓			✓	
ChangeStar [22]	✓				
DSAMNet [15]				✓	
P2V-CD [12]				✓	
ChangeFormer [1]	✓				
Changer [7]	✓				
TinyCD [5]	✓				
HANet [9]	✓				
LightCDNet [19]	✓				
CGNet [10]	✓				
BAN [11]	✓				
TTP [4]	✓				
MTKD [13]	✓				

and ChangeFormer, the improvements are significant. Notably, the original ChangeFormer customizes the configuration of MiT (e.g. depth, width of the network, etc.). In Open-CD, we restore the settings of b1~b5 in SegFormer [18] to make it scalable, and obtain higher F_1^c score with fewer parameters.

⁶https://github.com/likyoo/change_detection.pytorch

⁷<https://github.com/ChenHongruixuan/ChangeDetectionRepository>

⁸<https://github.com/Bobholamovic/CDLab>

⁹<https://github.com/Bobholamovic/ChangeDetectionToolbox>

5 Application

Algorithm implementation, validation and deployment. As mentioned above, Open-CD can be used for the implementation of custom change detection algorithms, validation of existing algorithms, and model deployment (relying on the OpenMMLab ecosystem, i.e., MMDeploy). In addition, it can also be applied as a construction tool for models, datasets, etc., promoting the exploration of semi-supervised and unsupervised algorithms.

Downstream validation for foundation model. Foundation models have attracted growing interest in remote sensing due to

Table 2: Results of different change detection methods on LEVIR-CD Test. † denotes the size of random crop is 512×512 , * denotes the batch size is set to 16, ‡ denotes the official code is based on Open-CD, and “300e” denotes 300 epochs. For BAN and TTP, only the learnable parameters are counted.

Method	Backbone	Lr Schd	Param (M)	GFLOPs	Inf (fps)	Precision ^c	Recall ^c	IoU ^c	F ₁ ^c	Official F ₁ ^c	Δ
FC-EF [6]	-	40k	1.353	3.244	66.37±0.0333	80.54	81.17	67.86	80.85	-	-
FC-Siam-Diff [6]	-	40k	4.385	1.352	46.29±0.0015	89.64	80.25	73.44	84.68	-	-
FC-Siam-Conc [6]	-	40k	4.989	1.548	45.67±0.0016	86.59	84.53	74.75	85.55	-	-
STANet-PAM [3]	ResNet-18	40k	13.356	48.083	1.53±0.0001	84.28	90.19	77.20	87.13	87.30	-0.17
DSIFN [20]	VGG-16	40k	35.995	78.982	5.19±0.0001	89.78	92.32	83.54	91.03	-	-
SNUNet-c16 [8]	-	40k	3.012	11.730	5.96±0.0001	93.02	89.75	84.09	91.36	-	-
BIT [2]	ResNet-18	40k	2.990	8.749	28.93±0.0006	93.17	88.38	83.00	90.71	89.31	+1.40
ChangeStar†* [22]	ResNet-18	40k	16.965	19.213	27.06±0.0007	93.75	88.90	83.92	91.26	90.29	+0.97
ChangeFormer-b0 [1]	MiT-b0	40k	3.847	2.455	25.26±0.0005	93.07	88.20	82.76	90.57	90.40	+0.17
ChangeFormer-b1 [1]	MiT-b1	40k	13.941	5.825	18.71±0.0004	93.09	89.26	83.71	91.14	90.40	+0.74
Changer†‡ [7]	ResNet-18	40k	11.391	5.955	49.31±0.2815	92.86	90.78	84.86	91.81	91.81	-
TinyCD [5]	-	40k	0.285	1.448	26.28±0.0004	91.87	89.89	83.26	90.87	91.05	-0.18
HANet [9]	-	40k	3.028	20.822	4.80±0.0001	91.72	89.09	82.46	90.39	90.28	+0.09
LightCDNet-large‡ [19]	-	40k	2.816	5.691	10.29±0.0002	92.68	89.67	83.74	91.15	91.43	-0.28
CGNet [10]	VGG-16	40k	38.989	87.550	1.93±0.0001	93.60	90.64	85.36	92.10	92.01	+0.09
BAN (ViT-L)†‡ [11]	MiT-b0	40k	4.474	307.374	1.87±0.0001	93.47	90.30	84.94	91.86	91.86	-
TTP†‡ [4]	ViT-L	300e	6.210	929.840	0.67±0.0001	93.00	91.70	85.60	92.10	92.10	-

their potential for generalization across tasks *e.g.*, object detection, semantic segmentation, and change detection. Open-CD offers a standardized and ready-to-use framework for evaluating change detection performance. Typically, MTP [17] uses a multi-task pre-trained encoder and evaluates it on change detection via Open-CD. **Downstream validation for generative model.** Although huge amount of remote sensing images are available through Google Earth *etc.*, data annotation remains a challenge, which is time-consuming and costly. One trend is to synthesize simulated image-label pairs through conditional GAN, Diffusion models, *etc.* [16, 21, 23] and use these data to (pre-)train downstream perception models. In this case, Open-CD can be applied to validate the quality of the simulated change detection data. A typical study is ChangeAnywhere [16], which proposes a conditional diffusion model to generate a simulated change detection dataset ChangeAnywhere-100K and uses Open-CD for subsequent validation.

Competition and challenge. As an advanced algorithm codebase, Open-CD can be easily used for change detection competitions and challenges. In general, users only need to customize their Dataset Class to enjoy all the advanced algorithms, data analysis tools, inference tools, *etc.* in Open-CD. Here, some solutions of winning competitions with Open-CD are listed:

- 2024 “Jilin-1” Cup.⁵ **2nd place.**⁶
- 2024 ISPRS TC 1 Contest.⁷ **Winner.**⁸
- 2023 “Jilin-1” Cup. **5th place.**⁹

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⁵<https://www.jl1mall.com/contest>

⁶<https://github.com/circleLZY/MTKD-CD>

⁷<https://www.gaofen-challenge.com/challenge>

⁸<https://github.com/DanyangLihhh/2024-ISPRS-TC-I-Contest>

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